

PRATT, KS, USA

WMO: 745430

Lat: 37.702N

Lon: 98.747W

Elev: 1952

StdP: 13.69

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 93978

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.1	11.6	-2.9	5.0	12.5	1.2	6.3	15.6	32.6	36.7	28.8	35.5	10.5	0	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	99.5	69.7	97.1	69.9	93.4	69.8	74.6	88.4	73.4	87.0	72.3	86.0	13.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.2	123.9	79.5	70.0	118.5	78.6	68.4	112.2	77.6	39.4	88.2	38.3	86.7	37.3	86.4	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.2	26.4	23.9	-1.2	103.3	6.2	3.2	-5.7	105.6	-9.3	107.4	-12.8	109.2	-17.3	111.5		
			-2.0	76.7	5.7	1.6	-6.1	77.8	-9.4	78.7	-12.6	79.6	-16.8	80.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	56.4	33.4	35.3	46.8	54.8	65.3	76.1	79.7	70.7	57.3	44.8	34.4		
		DBStd	18.86	10.42	11.08	10.26	9.71	8.82	5.91	5.93	6.06	7.52	9.43	8.98		
		HDD50	1908	517	423	179	59	4	0	0	0	37	204	485		
		HDD65	4672	978	833	567	326	108	3	1	0	30	271	607		
		CDD50	4261	4	10	81	204	478	782	920	850	620	263	47		
		CDD65	1547	0	0	3	20	117	335	456	385	199	32	0		
		CDH74	17780	1	3	59	319	1281	3844	5503	4251	2086	406	27		
		CDH80	8539	0	0	8	95	473	1852	2969	2146	876	117	3		
(14)	Wind	WSAvg	11.1	10.8	11.2	12.5	13.2	11.4	11.5	9.6	8.9	10.6	11.1	11.1	11.0	
(15)	Precipitation	PrecAvg	28.0	0.7	1.1	1.9	2.6	3.9	4.0	3.5	3.5	2.1	2.2	1.2	1.0	
(16)		PrecMax	37.9	2.2	2.9	6.7	5.7	9.7	8.3	6.6	7.9	5.6	6.8	3.6	3.5	
(17)		PrecMin	16.5	0.0	0.0	0.0	0.2	0.9	1.1	0.2	0.4	0.6	0.1	0.0	0.0	
(18)		PrecStd	5.3	0.6	0.9	1.6	1.4	2.0	1.9	1.7	2.1	1.2	1.7	1.1	1.0	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	68.3	72.0	80.9	89.8	95.0	101.6	104.0	103.8	98.5	88.8	78.5	63.4		
(20)		0.4% MCWB	48.3	50.7	56.9	60.9	64.2	68.7	69.8	69.8	68.1	64.5	55.6	47.4		
(21)		2% DB	60.7	65.2	73.6	81.6	89.8	96.6	99.8	99.2	92.9	82.4	72.0	58.9		
(22)		2% MCWB	45.3	50.3	56.7	59.8	64.7	69.8	69.9	70.8	68.4	62.6	54.5	45.3		
(23)		5% DB	54.5	59.0	70.1	77.1	85.6	93.1	97.3	95.2	89.6	78.9	64.6	54.2		
(24)		5% MCWB	42.8	46.2	54.9	59.3	65.3	69.1	70.5	70.6	68.1	60.4	51.1	44.2		
(25)		10% DB	49.9	53.6	64.0	72.3	81.3	90.1	93.5	91.0	85.5	73.0	60.7	50.0		
(26)		10% MCWB	40.7	43.3	52.2	57.2	64.3	69.5	70.7	70.3	66.5	59.6	49.8	41.8		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	51.3	55.4	63.8	66.2	72.1	75.5	76.4	76.3	73.5	70.5	59.3	53.3		
(28)		0.4% MCDB	60.9	61.7	72.9	77.6	81.5	88.9	89.2	91.0	85.0	80.7	68.4	56.4		
(29)		2% WB	46.4	50.8	60.7	63.4	69.9	73.4	74.6	74.4	71.6	67.4	56.5	47.7		
(30)		2% MCDB	58.4	62.8	68.5	73.8	81.1	87.7	88.1	88.7	83.3	76.3	66.5	56.2		
(31)		5% WB	43.4	47.2	57.8	61.2	68.3	72.2	73.3	73.1	70.3	64.1	53.4	44.5		
(32)		5% MCDB	54.1	59.4	65.9	71.9	79.6	86.2	87.0	87.0	82.0	73.5	62.9	53.0		
(33)		10% WB	40.7	43.3	53.6	58.9	66.6	70.9	72.2	71.8	68.9	61.0	50.7	41.6		
(34)		10% MCDB	49.0	52.6	63.1	70.6	77.7	84.4	86.4	85.6	80.6	70.9	59.5	49.6		
(35)	Mean Daily Temperature Range	MDBR	22.0	21.9	23.9	24.0	22.8	23.3	23.1	22.4	23.0	24.1	23.7	20.7		
(36)		5% DB MCDBR	30.8	32.9	31.4	31.5	28.1	27.2	28.0	28.2	26.9	30.8	31.6	28.0		
(37)		5% DB MCWBR	18.3	19.2	16.5	14.9	10.5	8.1	7.2	7.9	8.4	13.9	17.0	17.0		
(38)		5% WB MCDBR	29.4	30.8	24.4	26.2	22.6	23.8	23.0	23.4	22.3	23.6	27.1	25.4		
(39)		5% WB MCWBR	18.1	19.1	14.9	15.1	10.9	9.0	7.9	7.9	8.6	13.3	16.5	16.7		
(40)	Clear-Sky Solar Irradiance	taub	0.274	0.285	0.323	0.343	0.368	0.393	0.405	0.396	0.363	0.332	0.288	0.273		
(41)		taud	2.526	2.502	2.411	2.394	2.405	2.354	2.348	2.344	2.409	2.471	2.531	2.529		
(42)		Ebn at Noon	294	303	298	297	290	281	276	277	281	281	285	286		
(43)		Edn at Noon	24	28	34	37	37	39	39	38	34	29	24	23		
(44)	All-Sky Solar Radiation	RadAvg	826	1071	1417	1763	2017	2245	2199	1961	1630	1202	900	725		
(45)		RadStd	76	133	105	102	151	144	117	109	101	135	92	64		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page